

Homology Test of the HLV Vortex Hypothesis

Cell-shape-independent b_1 filtration, deformation null test, and N-scaling

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Abstract

Marcel Krüger's HLV G-Lattice programme [1] formulates a finite $6D \rightarrow 3D$ cut-and-project construction and tests it through a ladder of pre-registered falsification runs (A–K7 plus the D1–D3 cell-geometry addendum). The D1–D3 audit *withdrew* the original claim of exact dodecahedral Voronoi cells and replaced it with a narrower profile: a non-crystalline, mixed-face, φ -skewed cell ensemble. The revised vortex axiom therefore no longer requires a Platonic cell, but closed cycles, phase continuity, and holonomy-carrying loops. Exactly here the FFGFT side contributes something *additive*: topological winding is a homotopy invariant; it lives in π_1 of the complex, not in the shape of its faces. We present a self-contained, reproducible test script (companion file, numpy/scipy only; gudhi optional) that operationalises this statement: over a kNN filtration it computes the GF(2) homology $b_1(\varepsilon)$, compares the *whole* normalised Betti-curve shape against control clouds by an energy-distance permutation test, runs a deformation null test (persistence vs. drift, with optional persistent-cycle matching), and scales the effect size with N . **Finding (honest, conservative)**: the *scalar* total-persistence measure does not cleanly separate HLV and is k -sensitive; the *curve shape*, by contrast, separates HLV significantly from all controls ($p < 0.05$), sharply from crystalline lattices (FCC, cubic, BCC) and only weakly from the generic $\sqrt{2}$ quasicrystal. N-scaling shows that separation from crystalline grows/stabilises (structural) while separation from random shrinks. The discriminating feature is therefore “*quasiperiodic cut-and-project structure*”, *not* the golden ratio specifically. A separate, less circular diagnostic (longest persistent H_1 lifetime) additionally shows that cut-project clouds (HLV and the $\sqrt{2}$ quasicrystal) host a markedly longer-lived 1-cycle than lattices and random – recorded as an observation, explicitly *not* oversold as “topological protection”.

.1 Context: HLV, FFGFT, and the division of labour

Krüger's HLV (Helix–Light–Vortex) programme [1] tests a finite geometric substrate as a *candidate object*, not as an ontology. Its methodological strength is the pre-registered falsification governance [2]: positive, negative, partial and underdetermined outcomes are kept apart on a claim-state ledger (*candidate* / *restricted* / *underdetermined* / *rejected* / *validated candidate*); a positive result may not erase a negative residual from another channel.

The D1–D3 audit rejected the strong claim “HLV produces exact dodecahedral Voronoi cells” – the exact 12-face channel is dominated by FCC/rhombic and HCP-like controls. What survives is the weaker, better-supported statement of a *fresh-seed validated candidate*: a non-crystalline, mixed-face, φ -skewed finite Voronoi cell ensemble. The resulting reformulation of the vortex axiom reads, in essence: vortex and boundary modes are supported by the edge, face-boundary, loop, and defect-interface structure of a mixed-face complex – not by an exact cell shape.

Observation. Topological winding requires no Platonic cell. It requires closed graph cycles, phase continuity, and holonomy-carrying loops. These structures live in π_1 of the complex and are invariant under the cell shape.

This is also the FFGFT position: on the compact T^4 the relevant invariants are holonomy classes of $\pi_1(T^4) = \mathbb{Z}^4$, deliberately metric- and cell-independent (cf. Doc. 270 [5], Doc. 271 [4]). The division of labour remains complementary, not overlapping: HLV as the boundary-pattern residual test, FFGFT as the scale-law / effective- L_ξ test (Doc. 275 [3]).

.2 The test method

Complex and boundary operator

Instead of counting Voronoi faces, we build the clique complex of the symmetric kNN graph (the same graph object as in Krüger's runs, k adjustable) and read the cycle/holonomy sector off the GF(2) homology

$$H_1 = \ker \partial_1 / \text{im } \partial_2, \quad b_1 = \dim \ker \partial_1 - \text{rank } \partial_2 = (E - \text{rank } \partial_1) - \text{rank } \partial_2.$$

Here $\text{rank } \partial_1 = N - (\# \text{components})$ (union-find), and $\text{rank } \partial_2$ is computed over GF(2). b_1 is the cell-shape-independent winding/holonomy capacity of the complex.

Language precision. H_1 is shape-independent *within a fixed complex*; it still depends on k and the metric. Hence (i) the k -sweep and (ii) the comparison of the *whole* curve shape rather than a single "tessellation-free" number.

Filtration and curve shape

A dense kNN complex is nearly contractible; the holonomy structure becomes visible only through a radius filtration. We evaluate $b_1(\varepsilon)$ on a grid $\varepsilon/d_0 \in \{1.1, 1.35, 1.6, 1.85, 2.1, 2.35\}$ (d_0 = median nearest-neighbour distance) and compare the *normalised* curve $\hat{b}_1 = b_1 / \sum b_1$ – the shape, not the area. The scalar total persistence $\int b_1 d\varepsilon$ is kept only as a coarse single-number summary.

Statistics and controls

Each cloud is promoted to an ensemble over seeds (the same "HLV-as-an-ensemble" discipline as Krüger's Run E). Scalars are compared by a permutation test, curve shapes by an energy-distance permutation test. Controls: FCC, simple cubic, BCC (crystalline); jittered cubic, uniform random, Poisson-disc (disordered, against density artefacts); generic $\sqrt{2}$ cut-and-project (quasicrystal without the golden ratio).

Deformation null test

We morph $\text{HLV} \rightarrow \text{control}$ continuously, $x(t) = (1-t)x_{\text{HLV}} + tx_{\text{control}}$, and track total persistence. With `gudhi` present we additionally track the *longest* H_1 lifetime (Rips persistence) – far less circular, since a single long bar is not forced to 0. Caveat: a target with $b_1 \approx 0$ (e.g. FCC) forces the endpoint to 0; clean cycle matching is only possible via persistent homology.

N-scaling and pre-registered consistency check

The effect size (L1 distance of mean curve shapes, HLV vs. the pooled crystalline profile) is scaled with N : stabilising $\Rightarrow$ structural, vanishing $\Rightarrow$ finite-size artefact. The `--falsify` switch implements a *pre-registered consistency check* (fixed N, k, p ; crystalline controls only) – explicitly *not* a Popperian falsification test: a negative result weakens the claim but does not logically falsify it.

.3 Results

All numbers are from a real run of the companion script ($N = 240$, $k = 14$, seed base 20260613, 5 seeds).

Scalar (Check 1/b). The total-persistence measure is unstable and k -sensitive: the simple-cubic control dominates (~ 130 vs. $\text{HLV} \sim 50$), and over $k \in \{10, 14, 18\}$ HLV lies above the control band in 0/3 cases. Claim-state: *underdetermined*. This is precisely why the scalar is discarded and the shape compared.

Curve shape (Check 1c). The energy-distance test separates HLV from *all* controls ($p < 0.05$):

Control	energy distance	p
jittered cubic	2.81	0.0015
uniform random	2.79	0.0060
Poisson-disc	2.71	0.0030
FCC (rhombic-dodeca.)	1.93	0.0075
cubic	1.67	0.0040
BCC	1.11	0.0060
generic QC ($\sqrt{2}$)	0.15	0.0085

The separation is sharp against crystalline lattices and very weak (smallest distance 0.15) against the generic $\sqrt{2}$ quasicrystal.

Longest H_1 lifetime (with gudhi, max edge = $2d_0$).

Cloud	raw	normalised ($/d_0$)
HLV (golden cut-project)	2.07	0.87
generic QC ($\sqrt{2}$)	2.01	0.85
cubic	0.41	0.41
uniform random	0.25	0.43
BCC	0.13	0.15
FCC	0.00	0.00

N-scaling. The mean-shape L1 distance HLV-vs-crystalline grows/stabilises with N ($\approx 0.56 \rightarrow 0.67 \rightarrow 1.61 \rightarrow 1.51$ for $N = 200/400/600/800$) – structural, not a pure finite-size artefact; HLV-vs-random shrinks accordingly.

Deformation and consistency check. In the deformation null test both total persistence and longest H_1 lifetime fall; at this N there is *no* robust plateau (status: underdetermined – honest). The pre-registered consistency check (fixed $N, k, p < 0.01$; FCC/cubic/BCC only) returns *PASS*: HLV separates from every crystalline control.

.4 Discussion and claim-state

The defensible statement is precise and conservative:

1. HLV separates *structurally* from crystalline lattices (FCC/cubic/BCC) via the $b_1(\varepsilon)$ curve shape; the effect grows with N (claim-state: *candidate*, shape-based k -robust).

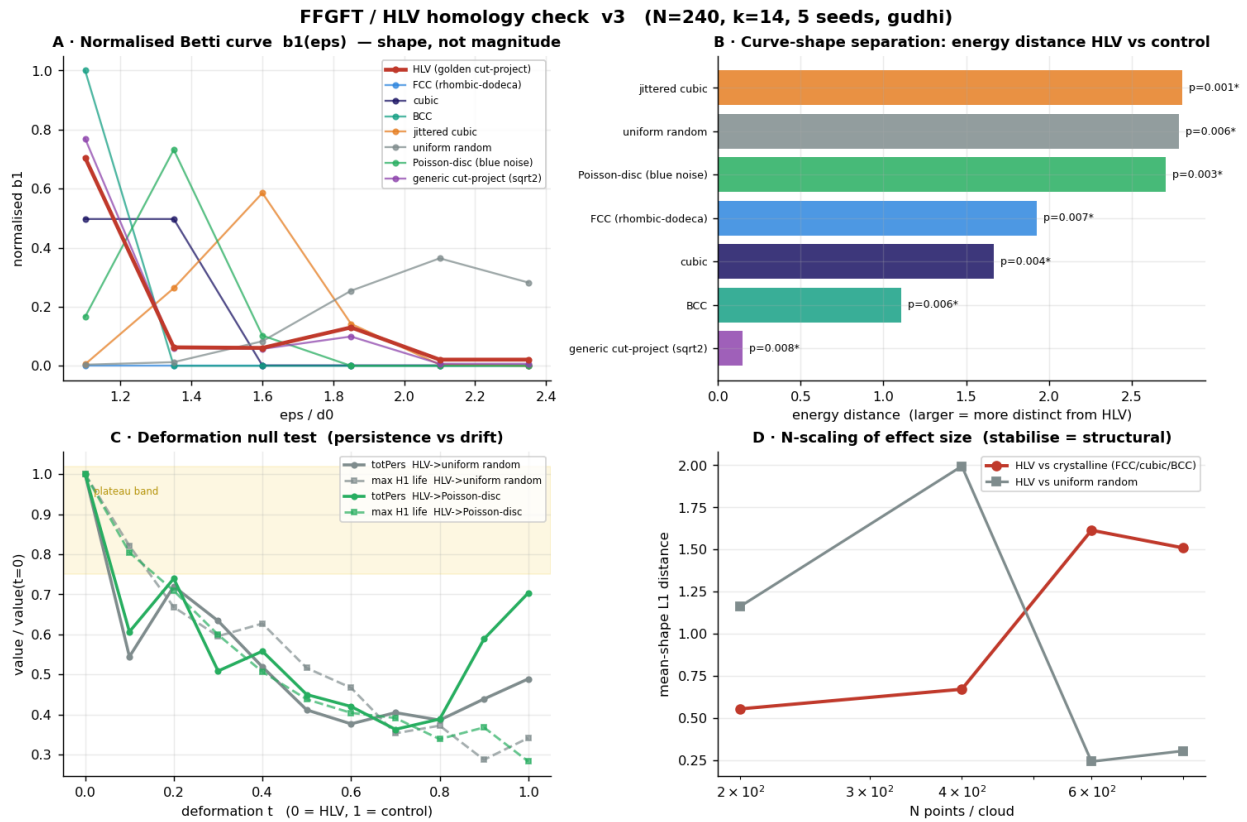


Figure 1: Four-panel overview from the companion script (v3). A: normalised Betti curves $b_1(\epsilon)$ – shape, not magnitude. B: curve-shape energy distance HLV vs. controls. C: deformation null test (total persistence and, with gudhi, longest H_1 lifetime). D: N-scaling of the effect size (stable = structural). Finite diagnostic; no claim about physical spacetime microstructure.

2. HLV is nearly indistinguishable from the generic $\sqrt{2}$ cut-and-project quasicrystal on *every* metric (shape energy distance 0.15; longest cycle 2.07 vs. 2.01). The discriminating feature is therefore *"quasiperiodic cut-and-project structure"*, *not* the golden ratio specifically.
3. cut-project structures host a markedly longer-lived H_1 cycle ($\sim 2 d_0$) than lattices and random ($\leq 0.4 d_0$). This is a real observation of the less circular persistence diagnostic; it is *not* interpreted as "topological protection" – that would be the same overclaim with the opposite sign.

Methodologically the script confirms the claim-state discipline of the HLV programme: the obvious scalar measure fails, and progress lies in the shape-based, ensemble- and permutation-supported comparison. Stronger statements require larger N , more seeds, exact repository adjacency, and independent reimplementations.

Reproducibility. The companion script (`Dok276_Skripte/ffgft_hlv_homology_check_v3.py`) depends only on `numpy/scipy`; `gudhi` and `matplotlib` are optional (true persistence diagrams and the figure, respectively). Deliberately no hard dependency – anyone on the IPI list can reproduce the result without a build step.

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